

PSTAT 126: REGRESSION ANALYSIS

Homework 2

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Section Time: Wednesday 5 PM

Question 1: Consider the simple linear regression model

$$Y = (\beta_0) + \beta_1 x + \epsilon$$

where the intercept β_0 is known.

1. Find the least-squares estimator of β_1 for this model.
2. What is the variance of the slope ($\hat{\beta}_1$) for the least-squares estimator found in the above part?

Consider the hypothesis test

$$H_0: \beta_1 = 0 \text{ vs. } H_1: \beta_1 \neq 0$$

3. What is the rejection region for the test?
4. What is the p -value of the test? (Bonus point)

$$1. \min \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = f \quad \frac{\partial f}{\partial \beta_1} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i) x_i = 0$$

$$\sum_{i=1}^n y_i x_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0 \quad \sum_{i=1}^n y_i x_i - \sum_{i=1}^n \beta_0 x_i - \sum_{i=1}^n \beta_1 x_i^2 = 0$$

$$\beta_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n \beta_0 x_i - y_i x_i \quad \boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (\beta_0 - y_i) x_i}{\sum_{i=1}^n x_i^2}}$$

$$2. \text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum_{i=1}^n (\beta_0 - y_i) x_i}{\sum_{i=1}^n x_i^2}\right) = \frac{1}{(\sum_{i=1}^n x_i^2)^2} \text{Var}\left(\sum_{i=1}^n x_i (\beta_0 - y_i)\right)$$

$$= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \left[\text{Var}\left(\sum_{i=1}^n x_i \beta_0\right) + \text{Var}\left(\sum_{i=1}^n y_i x_i\right) - \text{Cov}\left(\sum_{i=1}^n x_i \beta_0, \sum_{i=1}^n y_i x_i\right) \right]$$

$\Rightarrow x_i$ & β_0 are known, so their variances are 0. Additionally they are independent of y_i , so the covariance term is 0.

$$= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \left[\sum_{i=1}^n x_i^2 \text{Var}(y_i) + 2 \sum_{i \neq j} x_i x_j \text{Cov}(y_i, y_j) \right] \Rightarrow \text{all } y_i \text{'s are iid so } \text{Cov}(y_i, y_j) = 0$$

$$= \frac{1}{(\sum_{i=1}^n x_i^2)^2} \left[\sum_{i=1}^n x_i^2 \sigma^2 \right] = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n x_i^2)^2} = \boxed{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}$$

3. The rejection region is anything outside $[-z_{\alpha/2}, z_{\alpha/2}]$ if σ^2 is known, or $[-t_{n-1, \alpha/2}, t_{n-1, \alpha/2}]$ if σ^2 is unknown. Use $n-1$ as the degrees of freedom because we are estimating the value for β_1 .

$$4. Z = \frac{\hat{\beta}_1}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}} \Rightarrow p\text{-value} = 2 \cdot P(Z > |z|)$$

$$t = \frac{\hat{\beta}_1}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}} \Rightarrow p\text{-value} = 2 \cdot P(T_{n-1} > |t|)$$

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Question 2: Find the $(1 - \alpha)$ confidence interval for the average value of y at the point $x = x_0$, when σ^2 is known and unknown using two-sided hypothesis testing.

$$E[\hat{y}|x_0] = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

$$\text{Var}[\hat{y}|x_0] = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right], \quad \hat{\sigma}^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

if σ^2 known if σ^2 unknown

if known use $Z_{\alpha/2}$ as critical value

if unknown use $t_{n-2, \alpha/2}$ as critical value

$$CI = (\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm Z_{\alpha/2} \cdot \sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad \text{if } \sigma \text{ is known}$$

$$CI = (\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{n-2, \alpha/2} \cdot \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad \text{if } \sigma \text{ is unknown}$$

Question 3: Given the simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, 1)$ and are independent and identically distributed, find the maximum likelihood estimates of $\beta_0, \beta_1, \sigma^2$.

$\epsilon_i \sim \mathcal{N}(0, 1) \Rightarrow y_i \sim (\hat{\beta}_0 + \hat{\beta}_1 x_i, 1)$ by the assumption

$$\begin{aligned} L(y_i, x_i; \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_i - \mu}{\sigma}\right)^2\right) = \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \\ &= \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \prod_{i=1}^n \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) = \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \exp\left(-\sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \\ &= \prod_{i=1}^n \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right) \end{aligned}$$

plug in $\sigma^2 = 1 \Rightarrow \prod_{i=1}^n \left(\frac{1}{2\pi}\right)^{1/2} \cdot \exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right)$

$$\begin{aligned} \ell &= \ln(L(y_i, x_i; \beta_0, \beta_1)) = \ln\left(\prod_{i=1}^n \left(\frac{1}{2\pi}\right)^{1/2}\right) + \ln\left(\exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right)\right) \\ &= \sum_{i=1}^n \ln\left(\left(\frac{1}{2\pi}\right)^{1/2}\right) + \ln\left(\exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right)\right) \\ &= \sum_{i=1}^n \frac{1}{2} \ln\left(\frac{1}{2\pi}\right) + \ln\left(\exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right)\right) \\ &= -\frac{n}{2} \ln(2\pi) + \ln\left(\exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right)\right) \end{aligned}$$

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^n -(y_i - \beta_0 - \beta_1 x_i) = 0 \Rightarrow \text{same as } \hat{\beta}_0$$

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

$$\frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^n -(y_i - \beta_0 - \beta_1 x_i) x_i = 0 \Rightarrow \text{same as } \hat{\beta}_1$$

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

$\sigma^2 = 1$ by the consequence of the assumption that $\epsilon_i \sim \mathcal{N}(0, 1)$

Question 4:

1. Prove that the maximum likelihood estimate of σ^2 is biased.
2. Prove that the maximum likelihood estimate of σ^2 is asymptotically unbiased.
3. Determine the bias in the maximum likelihood estimate of σ^2 .
4. Prove that the variance of an estimate of σ^2 obtained using MLE method is less than the variance of an estimate of σ^2 obtained using OLS method.

1. $\hat{\sigma}^2 = \frac{SSE}{n}$, since MLE is not model dependent

$$\hat{\sigma}^2 = \frac{SSE}{n-2} \Rightarrow SSE = \hat{\sigma}^2 (n-2)$$

$$\tilde{\sigma}^2 = \left(\frac{n-2}{n}\right) \hat{\sigma}^2 \Rightarrow E[\tilde{\sigma}^2] = E\left[\left(\frac{n-2}{n}\right) \hat{\sigma}^2\right] = \frac{n-2}{n} E[\hat{\sigma}^2] = \frac{(n-2)}{n} \sigma^2$$

3. Bias($\tilde{\sigma}^2$) = $\frac{(n-2)}{n} \sigma^2 - \frac{n}{n} \sigma^2 = -\frac{2}{n} \sigma^2 \neq 0$
 $\Rightarrow \therefore \tilde{\sigma}^2$ is biased.

2. $\lim_{n \rightarrow \infty} E[\tilde{\sigma}^2] = \lim_{n \rightarrow \infty} \frac{n-2}{n} \sigma^2 = \sigma^2 \checkmark$

4. $Var(\tilde{\sigma}^2) = Var\left(\left(\frac{n-2}{n}\right) \hat{\sigma}^2\right) = \left(\frac{n-2}{n}\right)^2 Var(\hat{\sigma}^2)$

Since n is positive and presumably > 2 , $\left(\frac{n-2}{n}\right)^2 < 1$.
 This means $Var(\tilde{\sigma}^2)$ is a fraction of $Var(\hat{\sigma}^2)$,
 proving the variance of $\tilde{\sigma}^2$ is strictly less than $\hat{\sigma}^2$.

Question 5: Prove the following properties of β 's.

1. Use the fact that $\hat{\beta}_1$ and $\hat{\beta}_0$ are linear combination of y_i to find the distribution of $\hat{\beta}_1$ and $\hat{\beta}_0$.

$$2. \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}} \sim N(0, 1)$$

$$3. \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} \sim N(0, 1)$$

$$4. \frac{\hat{\beta}_0 - \beta_0}{\sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}} \sim t_{n-2}$$

$$5. \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{MSE}{s_{xx}}}} \sim t_{n-2}$$

1. $\hat{\beta}_1 = \sum_{i=1}^n k_i y_i \Rightarrow$ the sum of n y_i 's. Each $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ & the num of normal variables is a normal variable.

$$\therefore \hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{s_{xx}})$$

Similarly, since $\hat{\beta}_0$ is also a linear combination of y_i 's, the same rule applies

$$\Rightarrow \hat{\beta}_0 \sim N(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right))$$

$$3. \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} \Rightarrow \beta_1 = \mu_{\hat{\beta}_1} \Rightarrow \frac{\hat{\beta}_1 - \mu_{\hat{\beta}_1}}{sd_{\hat{\beta}_1}} \Rightarrow \text{standardizing a normal variable, so it is a standard normal variable.}$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} \sim Z = N(0, 1)$$

$$4. \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MSE/s_{xx}}} = \frac{\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/s_{xx}}}}{\frac{1}{\sqrt{\sigma^2/s_{xx}}}} = \frac{Z}{\sqrt{\frac{SSE/\sigma^2}{n-2}}} = \frac{Z}{\sqrt{\chi^2_{n-2}/(n-2)}} \sim t_{n-2}$$

$$5. \frac{\hat{\beta}_0 - \beta_0}{\sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}} = \frac{\frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}}}{\frac{1}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}}} = \frac{Z}{\sqrt{\frac{SSE/\sigma^2}{(n-2)}}} = \frac{Z}{\sqrt{\chi^2_{n-2}/(n-2)}} \sim t_{n-2}$$

$$2. \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}} \Rightarrow \frac{\text{value} - \text{mean}}{sd} \Rightarrow \text{standard normal}$$

$$\Rightarrow \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right)}} \sim Z = N(0, 1)$$

Question 6: Find the $(1 - \alpha)$ prediction interval for y at the point $x = x_0$, when σ^2 is known and unknown using two-sided hypothesis testing.

σ^2 is known $\Rightarrow$ use normal distribution

$$\begin{aligned} \text{prediction interval} \rightarrow \text{Var}(\widehat{y_i | x_0}) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) + \text{Var}(\epsilon_i) \\ &= \sigma^2 + \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \\ &= \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \end{aligned}$$

Critical value is $z_{\alpha/2}$

Prediction interval =

$$\begin{aligned} &\hat{y}_0 \pm z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \\ &= \left[\hat{y}_0 - z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \hat{y}_0 + z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \right] \end{aligned}$$

When σ^2 is unknown:

$$\text{Var}(\widehat{y_i | x_0}) = \text{MSE} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$\rightarrow$ use t distribution

critical value is $t_{n-2, \alpha/2}$

Prediction interval =

$$\begin{aligned} &\hat{y}_0 \pm t_{n-2, \alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \\ &= \left[\hat{y}_0 - t_{n-2, \alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \hat{y}_0 + t_{n-2, \alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \right] \end{aligned}$$

Question 7: Let $e_i := y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, then prove that

1. $\sum_{i=1}^n e_i = 0$

2. $\sum_{i=1}^n x_i e_i = 0$

3. $\sum_{i=1}^n \hat{y}_i e_i = 0$

4. $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$

5. The least-squares regression line always passes through the centroid $(\bar{x}, \bar{y})$ of the data.

$$\begin{aligned} 1. \sum_{i=1}^n e_i &= \sum_{i=1}^n y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = \sum_{i=1}^n y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i = \sum_{i=1}^n (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \\ &= \sum_{i=1}^n y_i - \bar{y} - \hat{\beta}_1 \sum_{i=1}^n x_i - \bar{x} = n\bar{y} - n\bar{y} - \hat{\beta}_1 (n\bar{x} - n\bar{x}) = 0 - \hat{\beta}_1 \cdot 0 \\ &= 0 \checkmark \end{aligned}$$

$$\begin{aligned} 2. \sum_{i=1}^n x_i e_i &= \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n x_i y_i - \hat{\beta}_0 \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2 \\ &= \sum_{i=1}^n x_i y_i - \bar{x} \sum_{i=1}^n y_i + \hat{\beta}_1 \bar{x} \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i (y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x}) \\ &= S_{xy} - \hat{\beta}_1 S_{xx} \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \\ &= S_{xy} - \frac{S_{xy}}{S_{xx}} \cdot S_{xx} = S_{xy} - S_{xy} = 0 \checkmark \end{aligned}$$

$$\begin{aligned} 3. \sum_{i=1}^n \hat{y}_i e_i &= 0 = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i) e_i = \sum_{i=1}^n \hat{\beta}_0 e_i + \sum_{i=1}^n \hat{\beta}_1 x_i e_i \\ &= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n x_i e_i = \hat{\beta}_0 \cdot 0 + \hat{\beta}_1 \cdot 0 = 0 \checkmark \end{aligned}$$

$$\begin{aligned} 4. e_i &= y_i - \hat{y}_i \Rightarrow y_i = \hat{y}_i + e_i \\ \sum_{i=1}^n y_i &= \sum_{i=1}^n \hat{y}_i + e_i = \sum_{i=1}^n \hat{y}_i + \sum_{i=1}^n e_i = \sum_{i=1}^n \hat{y}_i \checkmark \end{aligned}$$

$$\begin{aligned} 5. \sum_{i=1}^n e_i &= 0 \Rightarrow \sum_{i=1}^n y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = 0 \Rightarrow n\bar{y} - n\hat{\beta}_0 - n\bar{x}\hat{\beta}_1 = 0 \\ \Rightarrow \bar{y} - \hat{\beta}_0 - \bar{x}\hat{\beta}_1 &= 0 \Rightarrow \bar{y} = \hat{\beta}_0 + \bar{x}\hat{\beta}_1 \end{aligned}$$

$\Rightarrow$ shows that $(\bar{x}, \bar{y})$ is a point on the graph, regardless of what $n, \hat{\beta}_0, \hat{\beta}_1$ are.

CODING

1. Using the same dataset as stated in Homework 1, create a null model and a simple linear regression model. Plot the regression line passing through the points. Explain all the elements of the output obtained after calling the summary function.
2. Find the 95% confidence interval for all estimates and the 95% prediction interval.
3. Based on various measures such as residual standard error, coefficient of determination and F-value, comment on which model you would prefer.
4. Finally, interpret all the possible measures in the output, including the estimates, their standard errors, t-value, p-value, residual standard error, R^2 , F value.

In Rmd